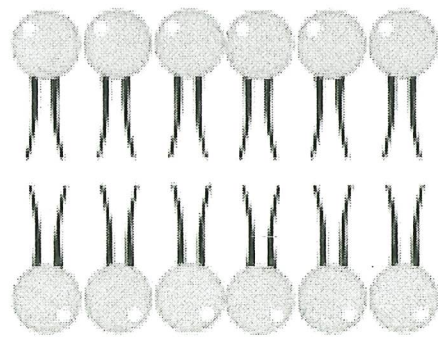
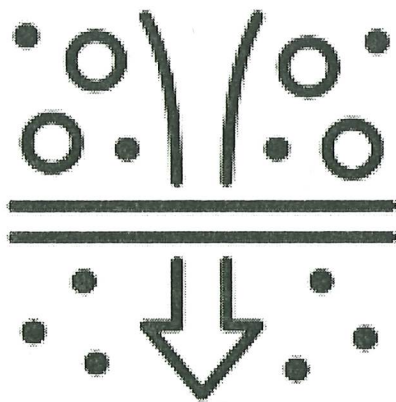


Unit 2 Part 1:

The Cell

Membrane and

Transport



Topics:

Cell Membrane Structure

Passive Cell Transport

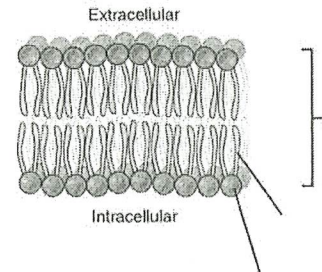
Active cell Transport

Cell Membrane

What are cells?

Cell Membrane

Label the cell membrane below

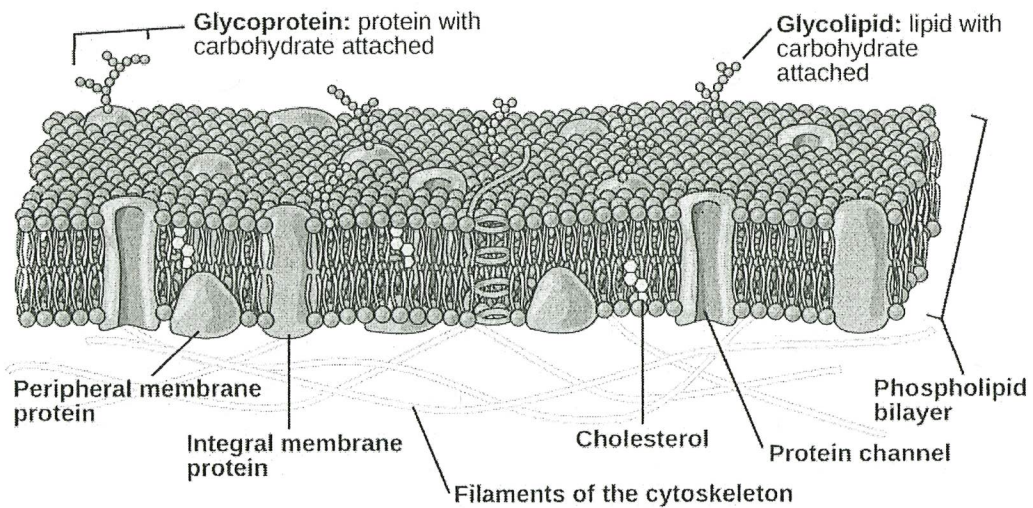


Why is the membrane flexible?

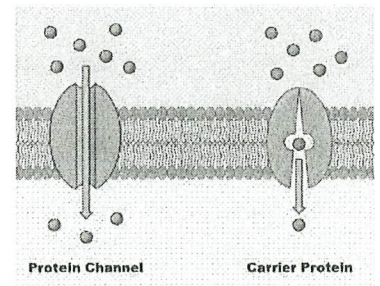
Cell Membrane Functions

1.	4.
2.	5.
3.	6.

How does the cell membrane maintain homeostasis?



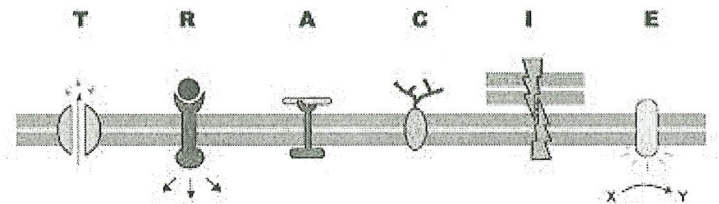
Integral Proteins



<u>Carrier</u>	<u>Channel</u>	<u>Receptor</u>

Membrane Jobs

T
R
A
C
I
E

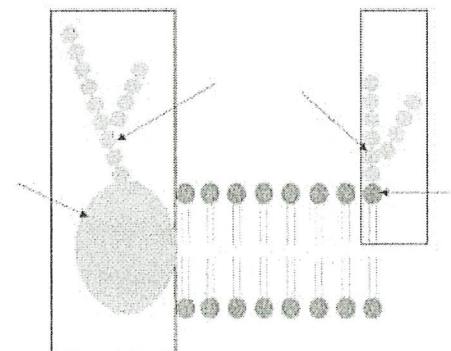


Peripheral Proteins

Cholesterol

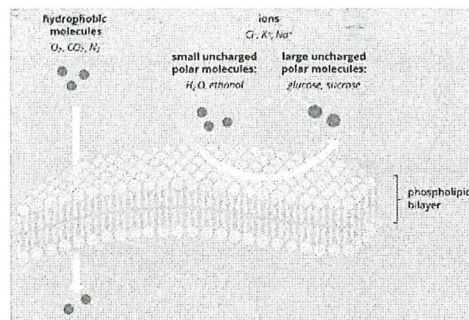
Label the Diagram Below

Oligosaccharides



Why is the cell membrane called a fluid mosaic model?

Selective Permeability

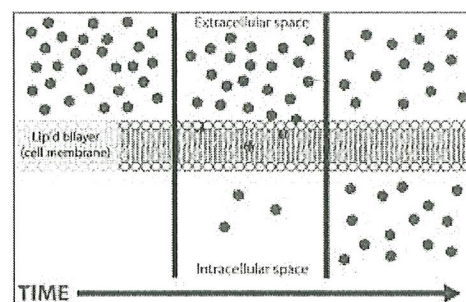


<u>Molecules that Pass Easily</u>	<u>Molecules that DO NOT Pass Easily</u>

Passive Cell Transport

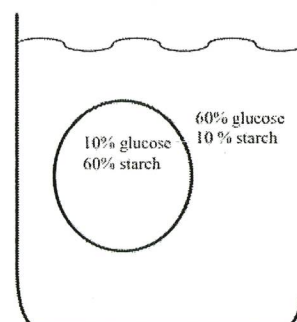
Passive Cell Transport

Simple Diffusion



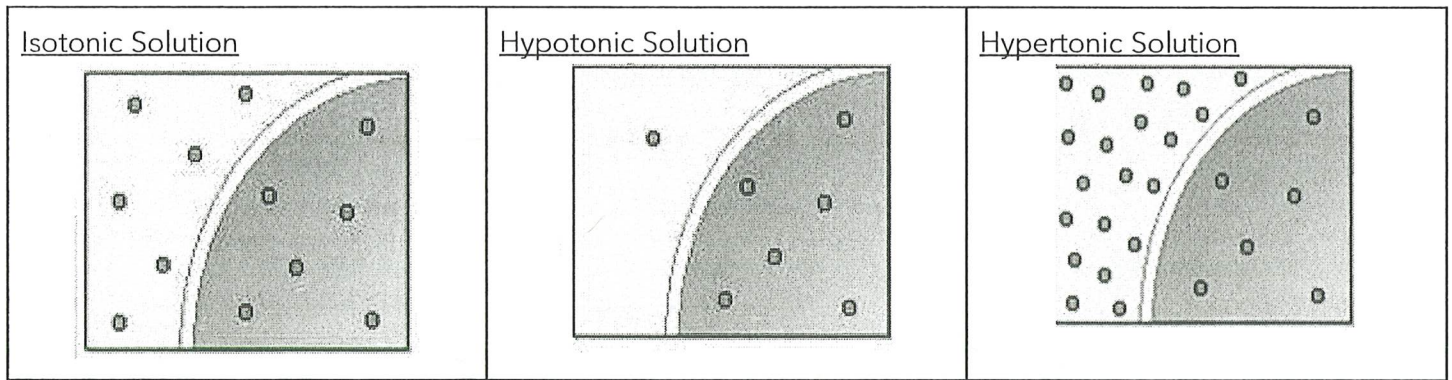
Practice- Diffusion

- The diagram below shows a cell in a solution. The cell's membrane is permeable to starch only.
- Which molecule will move?
- What direction will it move?



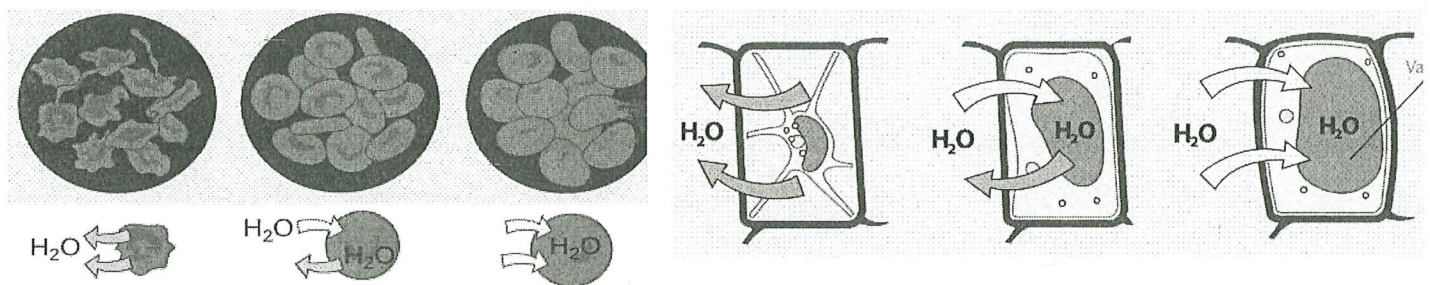
Osmosis

In each diagram below- show where the water will move by drawing arrows.



Osmosis in Animal Cells

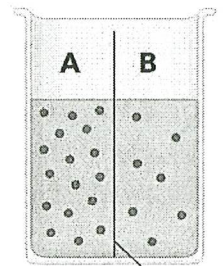
Osmosis in Plant Cells



Practice- Osmosis

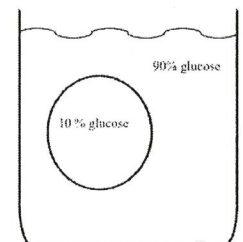
The beaker to the right is divided in half by a semipermeable membrane. Both sides are filled with water and sucrose molecules.

- Which side has the highest water potential?
- Which side has the highest solute concentration?
- In which direction will the water move?

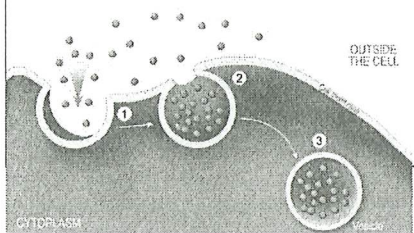


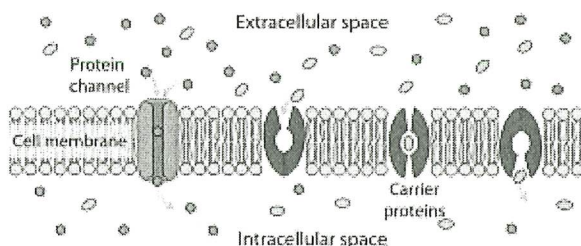
The cell to the right is placed in a solution of glucose. The cell membrane is only permeable to water.

- What type of solution is the cell surrounds the cell?
- Where will the water move?
- What will happen to the cell after the water moves?



Facilitated Diffusion

Channel Proteins	Carrier Proteins
ENDOCYTOSIS  OUTSIDE THE CELL CYTOPLASM	

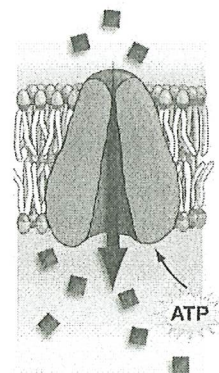


How does glucose enter the cell?

Active Transport

Active Cell Transport

Active transport

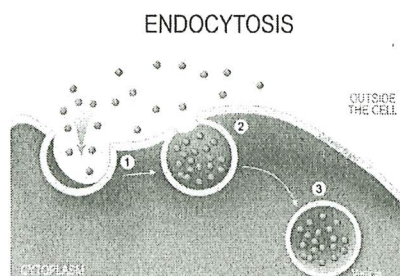
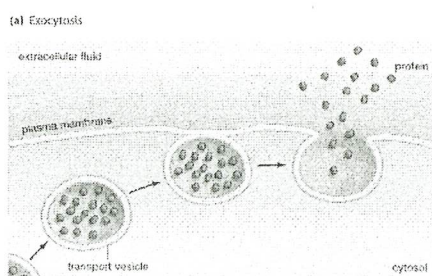


Molecules again move through a transport protein, but now energy must be expended to move them against their concentration gradient.

Exocytosis

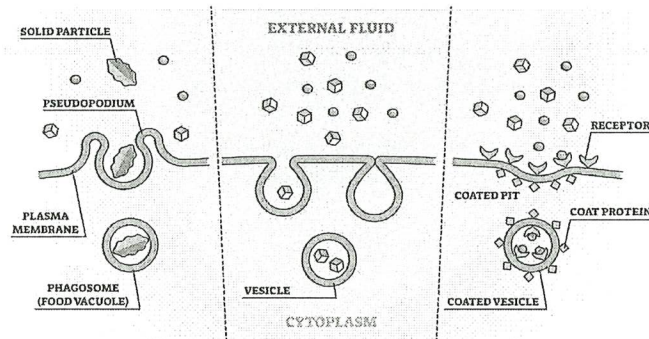
Endocytosis

Exocytosis Vs. Endocytosis

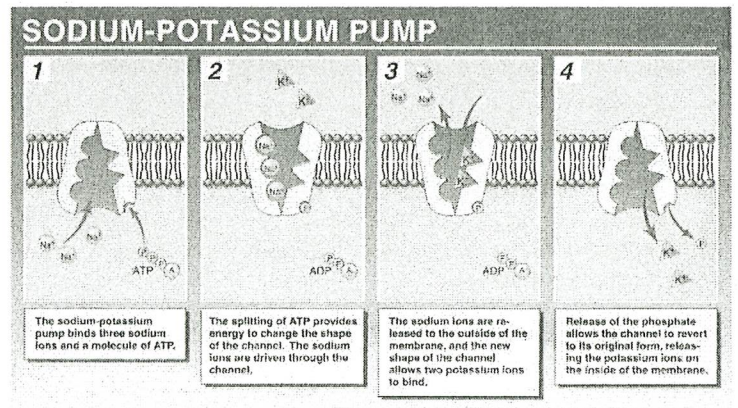


Pinocytosis	Phagocytosis	Receptor Mediated endocytosis

Label the types of endocytosis below



Sodium Potassium Pump



Cell Membrane – Structure and Function

Composition of the Cell Membrane and Functions

The **cell membrane** is also called the plasma membrane and is made of a phospholipid bilayer (A). Some of the functions of the cell membrane include protecting and enclosing the cell, giving shape to the cell, allowing transportation of materials in and out of the cell, and carrying out metabolic reactions near the inner surface of the cell membrane. The phospholipids of the cell membrane have a “hydrophilic” (water loving/attraction) head (B) and two “hydrophobic” (water fearing/repelling) tails (C). The head of the phospholipid is made of an alcohol and phosphate group, while the tails are chains of fatty acids.

The cell membrane is constantly vibrating, creating small openings within the structure. Therefore, the phospholipids can allow water and other smaller molecules to pass through into or out of the cell, without the use of energy. This type of passive transport is known as diffusion because the molecules are moving with the concentration gradient (high → low).

Another type of lipid in the cell membrane is cholesterol (D), which makes the membrane more fluid and adds to its flexibility. Embedded in the phospholipid bilayer are proteins that also aid in diffusion and cell recognition. Proteins called integral proteins (E) go all the way through the bilayer. Large molecules like glucose utilize these channel proteins to help move across cell membrane without the use of energy. Peripheral proteins (F) are a type of membrane protein that is permanently attached to the biological membrane. Some membrane proteins, called glycoproteins (G), have carbohydrate chains (H) attached to help cells recognize each other and certain molecules.

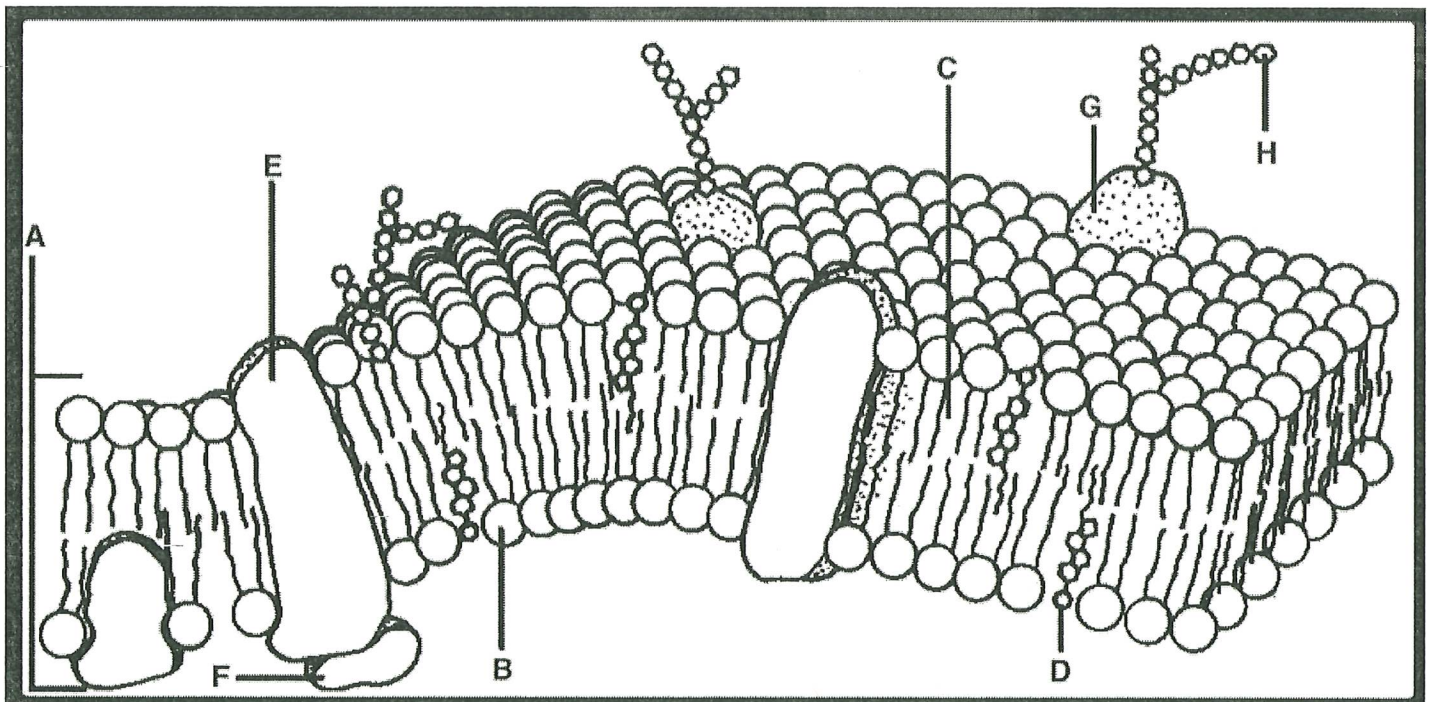
1. What is another name for the cell membrane? Identify the overall structure of the cell membrane.
2. Define hydrophilic. Which portion of the bilayer is hydrophilic?
3. Define hydrophobic. Which portion of the bilayer is hydrophobic?
4. When does the movement of materials in and out of the cell NOT require energy?
5. List the responsibilities of the cell membrane.
6. Describe the function of the proteins embedded in the cell membrane?
7. How are cells able to recognize various molecules and other types of cells?
8. What component(s) of the cell membrane contribute to the fluid quality of the cell membrane?

9. What is the role of carbohydrates on the surface of cell membranes?
10. How is cell-to-cell recognition important?
11. The cell membrane has glycoproteins and glycolipids. What do you think these words mean?

Cell Membrane Coloring

Correctly color code and identify the name for each part of the cell membrane.

Letter	Name/Color	Letter	Name/Color
_____	Phospholipid Bilayer (no color)	_____	Peripheral protein (red)
_____	Integral protein (purple)	_____	Cholesterol (brown)
_____	Fatty acid tails (orange)	_____	Carbohydrate chain (green)
_____	Phospholipid head (yellow)	_____	Glycoproteins (blue)



Match the cell membrane structure or its function with the correct letter from the diagram.

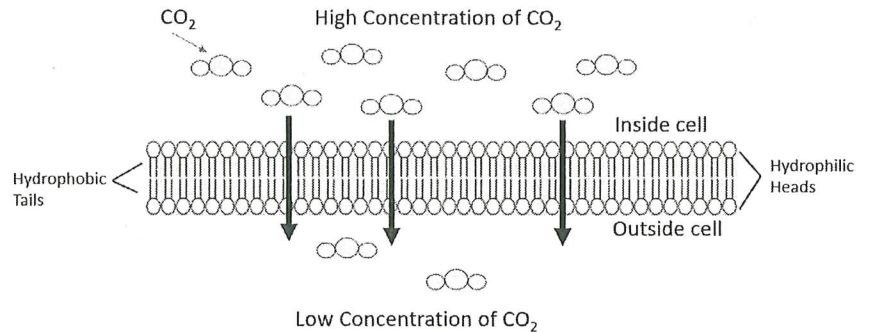
- | | |
|---|--|
| _____ Attracts water | _____ Repels water |
| _____ Helps maintain the flexibility of the membrane | _____ Makes up the bilayer (2 answers) |
| _____ Involved in cell-to-cell recognition | |
| _____ Helps transport materials, like glucose, across the cell membrane | |

Transport Drawing Practice

You will be determining how each molecule below moves through the cell membrane. Next, draw a diagram showing what this process looks like, labeling the parts of the cell membrane and molecules as needed. Use arrows, label the inside and the outside of the cell, etc. Add in membrane proteins as you need!

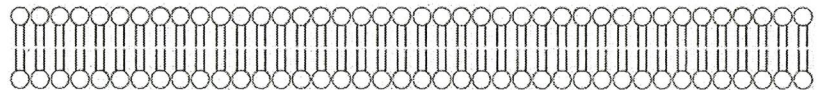
Example- Movement of CO₂ out of the cell

Type of Transport:	Method of Transport:
Passive	Simple Diffusion
Energy needed?:	Up or down the concentration gradient?:
No	Down



1. Movement of O₂ into the cell

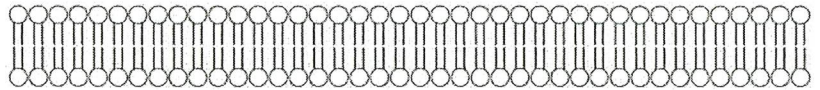
Type of Transport:	Method of Transport:
Energy needed?:	Up or down the concentration gradient?:



2. Movement of H₂O in a cell in a **hypertonic (high solute)** environment

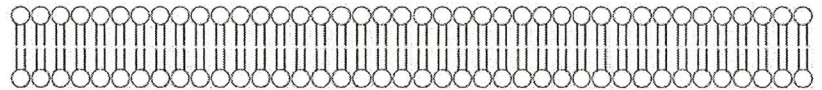
hypertonic solution → cell shrinks → cell loses water

Type of Transport:	Method of Transport:
Energy needed?:	Up or down the concentration gradient?:



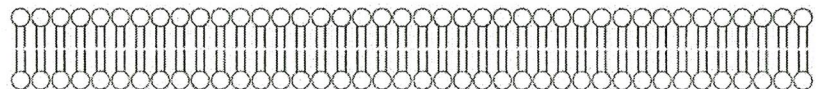
3. Movement of H₂O in a cell in a **hypotonic (low solute)** environment

Type of Transport:	Method of Transport:
Energy needed?:	Up or down the concentration gradient?:



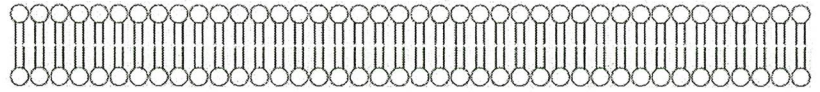
4. Glucose absorption

Type of Transport:	Method of Transport:
Energy needed?:	Up or down the concentration gradient?:



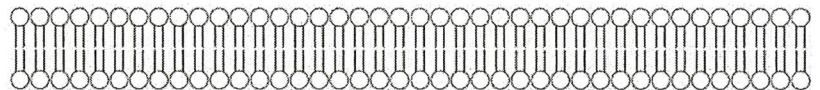
5. Hormone release

Type of Transport:	Method of Transport:
Energy needed?:	Up or down the concentration gradient?:



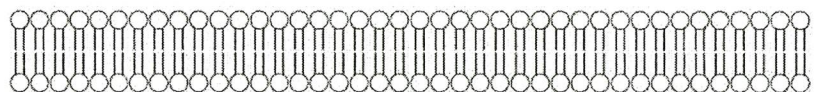
6. Engulfment of a bacteria

Type of Transport:	Method of Transport:
Energy needed?:	Up or down the concentration gradient?:

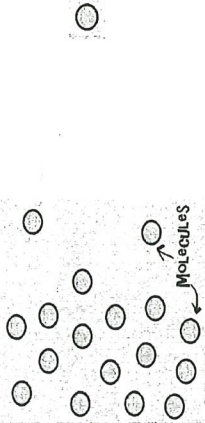


7. Movement of Na⁺ and K⁺

Type of Transport:	Method of Transport:
Energy needed?:	Up or down the concentration gradient?:



Amoeba Sisters Video Recap of Osmosis

1. The below picture represents diffusion of molecules. Place the following labels in the diagram: high concentration, low concentration, and an arrow showing the direction that the molecules would travel before equilibrium is reached. 	2. Osmosis is a type of diffusion, but it involves the movement of water. Similar to diffusion, osmosis is the movement of molecules (water molecules if osmosis) from a high concentration to a low concentration. The video clip explains that you can also look at water as moving to a _____ concentration of solute molecules. Why can it also be viewed this way? _____ _____ _____
3. Osmosis Scenario: The video clip mentioned a disaster scenario of a saltwater fish being placed in fresh water. What would occur if, instead, a freshwater fish was placed in saltwater? Your answer needs to have an arrow indicating the direction of water flow in osmosis, a label for "hypertonic," and a label for "hypotonic." 	4. Osmosis Scenario: Fluid movement into the brain after traumatic brain injury can result in dangerous brain swelling. One treatment that can be used in some of these cases is adding a _____ saline. You need to decide whether this blank should be the word hypertonic or hypotonic. Remember, you are trying to reduce the excessive fluid in the brain. Explain your answer: _____ _____ _____ _____

The Gummy Bear Mystery

Do you like gummy bears? We do! They are one of our favorite snacks, though we (try to) eat them in moderation because they are high in sugar. Consider that your sister is in a foul mood and decides to dump your gummy bears in your ice water about 30 minutes before you get home. The gummy bears are greatly enlarged by the time you get home! Your sister and some friends have different viewpoints for what happened. Draw a diagram below showing the enlarged gummy bears in a cup of water. Place the labels "hypertonic" and "hypotonic" in your diagram. One label should be for the gummy bears and one label should be for the water. 5.	Viewpoints: A) Your sister said that the sugar left the gummy bears, because the gummy bears were hypertonic compared to the water. B) Your friend Joe said that water traveled into the gummy bears, because the hypertonic compared to the water. C) Your friend Suzy said the sugar went into the gummy bears, because the gummy bears were hypotonic compared to the water. D) Your friend Will said that water traveled into the gummy bears by osmosis, because the gummy bears were hypotonic compared to the water.	6. Whose viewpoint is correct in the viewpoint column? A good answer has a good defense! Defend your answer, and also give reasons why the other explanations are incorrect. _____ _____ _____ _____ _____ _____ _____
---	---	--

Hypertonic, Hypotonic, or Isotonic? Oh My!

These red blood cells have all been placed in different solutions! Based on their appearance after being placed in these solutions for a period of time, place on each line (A) for hypertonic, (B) for hypotonic, or (C) for isotonic.

7. The cells are _____ compared to the solution.  SWELLING	8. The cells are _____ compared to the solution.  SHRINKING	9. The cells are _____ compared to the solution.  Stable
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Osmosis and Diffusion Lab

Why do red blood cells appear bigger after being exposed to distilled water?

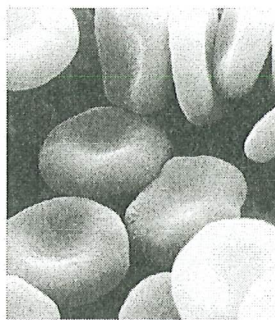
Introduction

All living things are made of **cells**. Some organisms, such as bacteria, are **unicellular**, which means they consist of a single cell. Other organisms such as humans, fish and plants are **multicellular**, which means they consist of multiple cells. All cells have some parts in common. One part found in all cells is the cell membrane. The **cell membrane** surrounds the cell, holds the other parts of the cell in place, and protects the cell. Molecules, such as oxygen, water, and carbon dioxide can pass in and out of the cell membrane. All cells also contain cytoplasm. The **cytoplasm** is a jelly-like substance inside the cell where most of the cell's activities are carried out. It's made out of water and other chemicals.

Some cells found in multicellular organisms are highly specialized and carry out very specific functions. An example of a specialized cell found in vertebrates is the **erythrocyte**, or **red blood cell (RBC)**. RBC's are by far the most abundant cells in the blood. The primary function of RBCs is to transport oxygen from the lungs to the cells of the body. In the capillaries, the oxygen is released so other cells can use it. Ninety-seven percent of the oxygen that is carried by the blood from the lungs is carried by hemoglobin; the other 3% is dissolved in the plasma. **Hemoglobin** allows the blood to transport 30-100 times more oxygen than could be dissolved in the plasma alone.

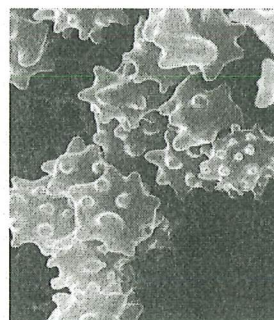
As you can see in the photo below, RBCs look like little discs when they are viewed under a microscope. They have no nucleus (to make room for more hemoglobin). A unique feature of RBCs is that they can change shape; this ability allows them to squeeze through capillaries without breaking. RBC's will also change shape in response to changes in the environment. For example, if you add a few drops of distilled water to blood on a microscope slide, the cells will look bigger after a few seconds.

Effects of tonicity in human red blood cells



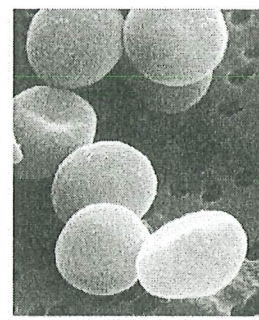
M. Shultz, R. Painter, and S. Singer, *Journal of Cell Biology*, 70:193 (1976) by permission of The Rockefeller University Press

A) Red blood cells immersed in an isotonic solution do not change in volume. The fluid portion of blood is normally isotonic with cytoplasm.



M. Shultz, R. Painter, and S. Singer, *Journal of Cell Biology*, 70:193 (1976) by permission of The Rockefeller University Press

B) Red blood cells immersed in a hypertonic solution shrivel up as water diffuses out of them.



M. Shultz, R. Painter, and S. Singer, *Journal of Cell Biology*, 70:193 (1976) by permission of The Rockefeller University Press

C) Red blood cells immersed in a hypotonic solution swell up as water diffuses into them.

Your Task

Carry out an experiment to determine what happens to red blood cells in different solutions.

Materials:

1. 3 gel deco cubes or gummy bears
2. 3 beakers
3. Electronic mass balance
4. ruler
5. Graduated cylinders or plastic cups
6. Tap Water
7. Various solutions
7. Distilled water

Procedure:

1. Obtain 3 deco cubes that have been soaking in tap water overnight. Put each one in a separate cup or beaker and label them 1, 2, and 3.
2. Measure the volume of each cube (see below) in cm and record your results in the data table.
3. Measure the weight of each cube in grams and record your results in the data table.
4. Submerge each cube in 3 solutions of your choice.
5. Leave cubes for at least 24 hours.
6. Repeat the measurements above for the cubes.

Pre-Lab Questions:

Answer the following questions to help you set up the experiment. Have your teacher check your setup (questions 1-10) before you begin your experiment.

1. What will be your **independent variable**? (Reminder: This is what you are changing across groups.)
2. What will be your **dependent variable**? (Reminder: This is what you are measuring.)
3. Which group will serve as your **control** in your experiment?
4. How will you ensure that your data is **high quality, and reduce error** as much as possible?
5. What **type of graph** will you use to display and interpret your data?

Data Table:

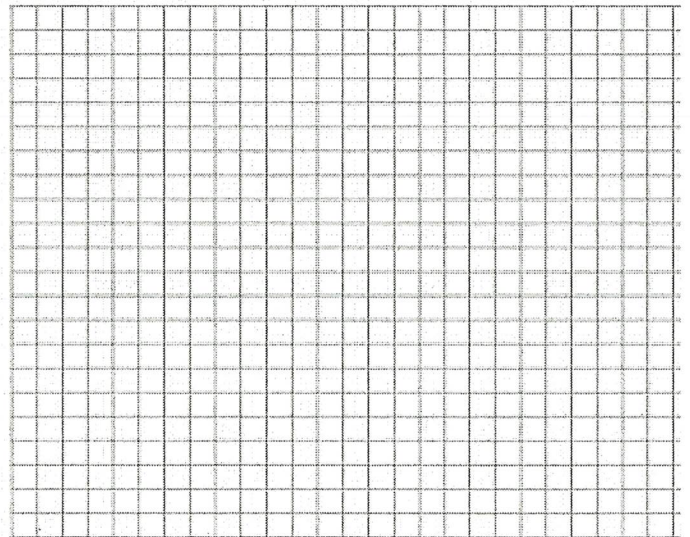
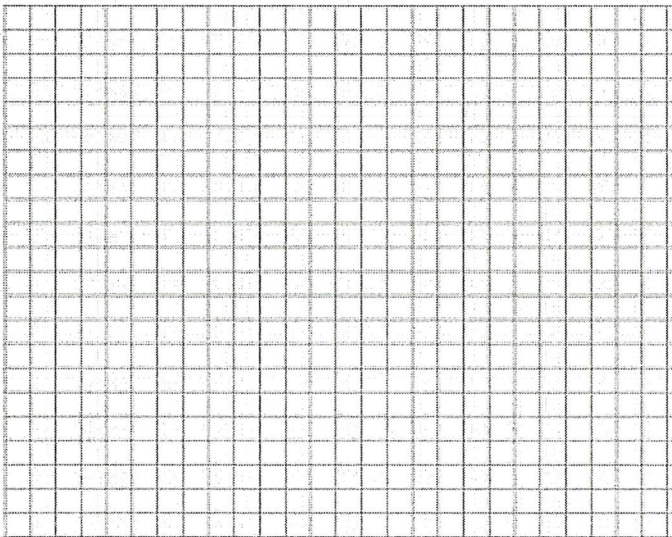
	Cube 1	Cube 2	Cube 3
Solution Type			
Mass Before Treatment (g)			
Mass After Treatment (g)			
Volume Before Treatment (cm ³)			
Volume After Treatment (cm ³)			

Post Lab Analysis Questions:

1. What does the **cube** represent in this experiment?
2. What substance is **actually** passing through the membrane? Explain.
3. What does a **positive** change in mass or volume actually represent? (Hint: what is causing the change?)

4. What does a **negative** change in mass or volume actually represent? (Hint: what is causing the change?)
5. If a cell is placed in a **hypertonic** environment, should it have a positive or negative change in mass? **Explain your answer.**
6. If a cell is placed in a **hypotonic** environment, should it have a positive or negative change in mass? **Explain your answer.**
7. Predict the change in mass of a cell placed in an isotonic environment. **Explain your answer.**

Graph your data on the graphs below. Make sure to use proper scale, units, and labels.



A Case Study on Water: Can too much of a good thing be deadly?

Jennifer Strange was a 28-year-old mother of three who entered a radio contest to try to win a Nintendo Wii game system for her children. As part of KDND's "Hold your wee for a Wii" contest, Jennifer drank bottle after bottle of water for hours without urinating. Initially, Jennifer seemed to be having fun, joking lightheartedly with the radio hosts and obligingly downing an **8-ounce bottle of water every 15 minutes**. As the hours passed, however, she developed a splitting headache and dizziness. Finally, she couldn't take it anymore and ran to the bathroom and vomited. Jennifer called her boss to say that she was going home for the day because her headache was excruciating and she was too sick for work. Jennifer's mother found her that afternoon, dead in her home.

James McBride, a 25-year-old police officer, died after a 12-mile bike ride that was part of a training course. Over the course of the ride, James drank **roughly three gallons of water** that he brought with him in a pack. During a session that focused on how to dismount a bike, an instructor noticed that McBride looked ill and asked him to sit down. McBride complained of dizziness and nausea, and then vomited. Officers initially thought he might have suffered heat stroke. Unaware that James had already consumed so much water, they gave him more water to cool him down while he sat out of the exercise. When another officer hurt himself during the exercise, an ambulance was summoned. The paramedics noticed that James was convulsing and continuing to vomit, so they brought him to the hospital. He died in the hospital the next day.

Directions: After reading the stories above, complete the CER below. Use your data from the lab to back up your claim. See the descriptions below on the parts of a CER. Use COMPLETE sentences and be thorough.

Driving Question: <i>Can too much water be deadly?</i>	
Claim:	
Evidence (from lab + case study):	Reasoning:

Cell Transport Situation Cards

For each situation, write what **type of transport** (passive or active), and then specify the **specific form of transport** (diffusion, exocytosis, endocytosis (pinocytosis), etc.). **If it is an OSMOTIC situation ONLY**, specify the **movement of water (in/out/both)** of the cell, whether the cell is in a **hypo/hyper/isotonic environment**, and what will happen to the cell (swell/shrink/equilibrium).

Scenario #	Type of Transport (A or P)	Form of Transport - be as specific as you can e.g. "endocytosis (phagocytosis)"	Water moving in or out? Hypo/hyper, or isotonic solution? (OSMOSIS ONLY)
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			

Cell Transport Review Questions

Part 1: Read the paragraph below and use your knowledge of osmosis to answer the questions.

Fertilizer was applied to three healthy tomato plants. Plant X was given 0.9 grams of fertilizer, plant Y was given 4.8 grams, and plant Z was given 18 grams of fertilizer. Plant X continues to be healthy and produce tomatoes. Plant Y wilted, but with continued watering managed to regain some of its former health. Plant Z died.

Was plant X given a hypertonic solution? Explain.

1. Did plant Y go through plasmolysis? Explain.
2. Did plant Y lose turgor pressure? Explain.
3. Was plant Z given an isotonic solution of fertilizer? Explain.
4. Did plant Z's cells burst due to cytolysis? Explain.

Part 2: Fill in the Blank.

5. A membrane that allows only certain substances to pass through is: _____
6. Name 3 examples of homeostasis. _____
7. When turgor pressure is lost, the cell contents shrink. What is this condition called?

8. The odor of BBQ chicken coming through the house is an example of: _____
9. Name 3 things that would affect the rate of diffusion through a membrane:

10. Osmosis refers to the diffusion of: _____
11. The engulfing process by which large molecules enter the cell is: _____
12. Organelles that push extra water out through the cell membrane are called:

13. The diffusion of molecules across a membrane through transport proteins is called:

14. The difference in concentration from one area to another is called a: _____
15. If a substance is too large to cross through the membrane, a cell may transport the substance in a:

Part 3: Match the Items. You will only use each word once.

Active transport	Permeable	Isotonic
Hypotonic	Osmosis	Diffusion
Homeostasis	Plasmolysis	Exocytosis
Turgor	Cytolysis	Contractile vacuole

16. _____ shrinking the cell content due to loss of water
17. _____ "steady state"
18. _____ movement of materials against diffusion pressure
19. _____ spreading out of molecules in a given space
20. _____ solution that contains a higher concentration of water molecules than the cell content
21. _____ passage of large molecules through the plasma membrane out of the cell

- 22. _____ diffusion of water through a selectively permeable membrane
- 23. _____ membrane through which substances can pass
- 24. _____ bursting of cells due to internal pressure

Part 4: Answer the Following Questions.

- 25. Explain 2 ways that large molecules can enter the cell through the cell membrane.
- 26. Describe a problem that marine saltwater organisms would have in fresh water.
- 27. Organize the terms isotonic, hypertonic, and hypotonic in order from the solution with the lowest concentration of dissolved particles to the highest concentration.
- 28. Suppose you have a container divided by a membrane that is permeable to water but not sugar. Side A has a 10% sugar solution. Side B has a 40% sugar solution. Both containers start out with water levels at 10cm in height. Over time, the height of one side drops to 7cm, and the height of the other side increases to 13cm. Which side of the container is now at 7cm? Explain.
- 29. Draw a phospholipid below. Label the hydrophobic and hydrophilic portions of the phospholipids.
- 30. Use arrows to show the direction of water movement into or out of each cell. Color and label the cell in an isotonic environment light blue, the hypotonic environment yellow, and the hypertonic environment light green.

